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import streamlit as st
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
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# =====:
# KONFIGURASI HALAMAN
# =====:
st.set_page_config(page_title="Dashboard Analisis Siswa",
st.title("📊 Dashboard Analisis Data Simulasi Siswa")
st.markdown("Analisis statistik dan segmentasi berbasis :
```

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DeltaGenerator()
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# =====:
# UPLOAD FILE (AMAN UNTUK STREAMLIT CLOUD)
# =====:
uploaded_file = st.file_uploader("Upload File Excel", ty

if uploaded_file is not None:

    df = pd.read_excel(uploaded_file)

    # Ambil semua kolom numerik
    indikator = df.select_dtypes(include="number")

    st.success("File berhasil dibaca!")

    # =====:
    # KPI Rata-rata Nilai
    # =====:
    mean_scores = indikator.mean()
    rata2_total = mean_scores.mean()

    col1, col2, col3 = st.columns(3)
    col1.metric("📈 Rata-rata Keseluruhan", f"{rata2_tot
    col2.metric("📊 Jumlah Soal", indikator.shape[1])
    col3.metric("📊 Jumlah Siswa", indikator.shape[0])
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col3.metric( 🧑 Jumián Siswa , indikator.srape[0])

st.divider()

# =====
# ANALISIS RATA-RATA PER SOAL
# =====
st.header(" 📊 Rata-rata Nilai per Soal")

fig_bar, ax_bar = plt.subplots(figsize=(8,4))
ax_bar.bar(mean_scores.index, mean_scores.values)
ax_bar.set_ylabel("Rata-rata")
ax_bar.set_xticklabels(mean_scores.index, rotation=45)
st.pyplot(fig_bar)

st.divider()

# =====
# HEATMAP KORELASI
# =====
st.header(" 📈 Korelasi Antar Soal")

corr = indikator.corr()

fig_corr, ax_corr = plt.subplots(figsize=(6,5))
im = ax_corr.imshow(corr, cmap="coolwarm", vmin=-1, vmax=1)
plt.colorbar(im, ax=ax_corr)

ax_corr.set_xticks(range(len(corr.columns)))
ax_corr.set_yticks(range(len(corr.columns)))
ax_corr.set_xticklabels(corr.columns, rotation=45, ha='right')
ax_corr.set_yticklabels(corr.columns)

st.pyplot(fig_corr)

st.divider()

# =====
# ANALISIS REGRESI (Soal Terakhir sebagai Target)
# =====
st.header(" 📉 Analisis Regresi Linear")

X = sm.add_constant(indikator.iloc[:, :-1])
y = indikator.iloc[:, -1]

model = sm.OLS(y, X).fit()

coef = model.params[1:]
r2 = model.rsquared

fig_reg, ax_reg = plt.subplots(figsize=(8,4))
ax_reg.bar(coef.index, coef.values)

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ax_reg.axhline(0, linestyle="--")
ax_reg.set_title("Koefisien Regresi")
ax_reg.set_xticklabels(coef.index, rotation=45)

st.pyplot(fig_reg)
st.info(f"Nilai R2: {r2:.3f}")

st.divider()

# =====
# SEGMENTASI K-MEANS
# =====
st.header("🎯 Segmentasi Siswa (Clustering)")

scaler = StandardScaler()
X_scaled = scaler.fit_transform(indikator)

kmeans = KMeans(n_clusters=3, random_state=42, n_init=1000)
cluster = kmeans.fit_predict(X_scaled)

indikator_cluster = indikator.copy()
indikator_cluster["Cluster"] = cluster

cluster_mean = indikator_cluster.groupby("Cluster").r

st.subheader("Rata-rata Tiap Cluster")
st.dataframe(cluster_mean)

# Radar Chart
labels = indikator.columns.tolist()
angles = np.linspace(0, 2*np.pi, len(labels), endpoint=False)
angles += angles[:1]

fig_rad = plt.figure(figsize=(6,6))
ax_rad = plt.subplot(polar=True)

for i, row in cluster_mean.iterrows():
    values = row.tolist()
    values += values[:1]
    ax_rad.plot(angles, values, label=f"Cluster {i}")
    ax_rad.fill(angles, values, alpha=0.2)

ax_rad.set_thetagrids(np.degrees(angles[:-1]), labels)
ax_rad.set_ylim(0, indikator.max().max())
ax_rad.set_title("Radar Segmentasi Siswa")
ax_rad.legend(loc="upper right")

st.pyplot(fig_rad)

st.success("Dashboard berhasil dijalankan 🎉")

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else:

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st.warning("Silakan upload file Excel terlebih dahulu")
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